

Bridging the Data Gap: Towards Infrastructure for Custom, Dynamic Dataset Curation & Analysis

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ABSTRACT

In the age of generative artificial intelligence (AI), data is at the center of our ability to adequately and sustainably develop advanced technologies that can support diverse end users. While for some populations and data types it is fairly easy to find usable, representative datasets and sources, it becomes challenging when the focus is on historically marginalized communities. This makes it difficult for research like our own, which aims to study media coverage, social discourse, and community-specific narratives to improve the cultural alignment of generative AI technologies. Existing news and media corpora aggregate hundreds of outlets but systematically under- or misrepresent niche and community-centered perspectives, presenting a gap in both existing data and dataset infrastructures. To fill this gap, and in alignment with our ongoing efforts to support

cultural alignment of generative AI technologies, we present a modular, open-source pipeline that continuously scrapes, normalises, and thematically annotates articles from seven prominent Black media publications on a rolling 30-day horizon. The pipeline delivers approximately 1,200 articles per monthly cycle, each tagged against a curated taxonomy of twelve social conflict topics using a hybrid TF-IDF + seed-phrase matcher, and exports research-ready CSV artefacts alongside a styled PDF editorial report, fully automated via GitHub Actions and deposited to Google Drive. We describe the pipeline architecture, dataset schema, quality metrics, and the design decisions that make the system extensible to other under-represented publication and data ecosystems.

CCS Concepts: • *Information systems~Data extraction and integration*; • *Information systems~Information retrieval*; • *Computing methodologies~Natural language processing*.

Keywords: dataset curation, media monitoring, Black press, topic modeling, TF-IDF, automated pipeline, social conflict, GitHub Actions

1 INTRODUCTION

Computational research on media framing, agenda setting, and coverage equity relies on large, timely corpora [1, 2]. Yet the datasets most commonly used in NLP and computational journalism research—AllTheNews, CC-News, RealNews, GDELT—are aggregated from a small set of legacy outlets whose editorial diversity is limited [3]. Publications that serve marginalised or community-specific audiences are almost entirely absent from these corpora.

The Black press is a historically significant and demographically distinct media ecosystem. Outlets such as *The Root*, *The Grio*, *Ebony*, *Essence*, *NewsOne*, *Capital B News*, and *Blavity* collectively publish hundreds of articles per month on politics, justice, health, culture, and economics—often foregrounding perspectives that mainstream outlets suppress or marginalise [4, 5]. Despite their importance for studying how racialised communities narrate social conflict, these outlets are largely absent from existing research corpora.

This paper presents a *live, configurable, open-source pipeline* designed to close that gap. The system scrapes the seven publications listed above on a rolling 30-day schedule, normalises each article to a common schema, applies a hybrid TF-IDF + seed-phrase topic tagger across twelve social conflict topic classes, and exports analysis-ready datasets alongside a styled research report—all without manual intervention. Our contributions are:

1. A reproducible, schema-documented dataset of Black media articles updated monthly via CI/CD automation.

2. A hybrid topic-annotation layer (TF-IDF keyword extraction + seed-phrase matching) extensible to arbitrary topic taxonomies.
3. An end-to-end reference architecture demonstrating how custom, under-represented media ecosystems can be continuously monitored with modest infrastructure costs.

2 BACKGROUND & MOTIVATION

2.1 Limitations of General-Purpose News Corpora

Large open news corpora provide convenient baselines for NLP benchmarking but exhibit well-documented source-diversity problems. The GDELT project [6] indexes millions of articles globally but weights towards high-Alexa-rank outlets. AllTheNews [7] and CC-News [8] employ similar crawl-frequency heuristics that de-prioritise low-traffic domains. Specialised datasets—MedMCQA, LegalBench, FinancialPhraseBank—address vertical domain gaps but have no analogue for community or ethnic press coverage.

2.2 The Black Press as a Research Target

The seven outlets studied here reach a combined monthly unique audience estimated at 35–50 million visitors [9]. Their coverage of policing, voter access, housing, maternal health, and reparations frequently differs in framing, sourcing, and intensity from mainstream coverage of the same events [10]. Studying these differences requires a corpus that is both timely and continuously updated, because the window of relevance for social conflict coverage is typically measured in weeks, not years.

2.3 Gap in Automated, Configurable Monitoring Tools

Existing media-monitoring services (e.g., Meltwater, Cision, GDELT Analysis Service) are proprietary, costly, and not designed to export clean research datasets. Open alternatives such as NewsAPI or MediaCloud provide useful APIs but do not index the Black press at sufficient depth or recency. No open-source, self-hosted tool exists that (a) targets this specific publication set, (b) applies a sociologically grounded topic taxonomy, and (c) runs fully automated on free CI/CD infrastructure.

3 SYSTEM ARCHITECTURE & APPROACH

The pipeline consists of four decoupled stages executed sequentially by a single orchestrator (run.py): **Scrape** → **Analyse** → **Report** → **Deliver**. Each stage reads from and writes to the outputs/ directory, enabling any stage to be re-run independently.

3.1 Stage 1: Scraping

Each of the seven publications requires a distinct scraping strategy due to differences in technology stack (Table 1).

Publication	Method	Fallback
thegrio.com	WP REST API	RSS
theroot.com	WP REST API	RSS
newsone.com	WP REST API	RSS
ebony.com	WP REST API	RSS
capitalbnews.org	Sitemap + HTML	RSS
essence.com	Paginated HTML	RSS
blavity.com	Playwright + HTML	RSS

Table 1: Per-publication scrape methods.

WordPress REST API. Four outlets expose `/wp-json/wp/v2/posts`. We issue paginated requests with after/before date filters and embed term and author metadata in a single round trip via `_embed=wp:term,wp:author`.

Sitemap + HTML (Capital B News). Capital B’s sitemap index exposes post, article, and news sub-sitemaps with `<lastmod>` timestamps. We parse these first (fast pre-filter), then discover additional URLs via RSS feed, category pages (up to 3 pages), and WordPress date archives. Candidates are capped at 200 to keep runtime under 5 minutes.

Playwright headless Chromium (Blavity). Blavity is a JavaScript-rendered SPA; its RSS feed is capped at 10 items. We use Playwright to navigate eight category pages (up to 10 pagination steps each), intercept article links, and extract metadata including JSON-LD structured data for publication dates. A requests-based HTML fallback is available if Playwright is unavailable.

All scrapers share a common normalisation step and a polite crawl delay of 250–500 ms between requests. A unified RSS fallback triggers when any primary method returns fewer than five articles.

3.2 Stage 2: Analysis

TF-IDF Keyword Extraction. We concatenate title and description for every article and vectorise the corpus using scikit-learn’s `TfidfVectorizer` with `n-gram` range (1,3), sublinear TF scaling, and NLTK English stopwords augmented with a domain-specific extension list. The top-*k* terms by aggregate TF-IDF weight are reported as the period’s “signal keywords.”

Seed-Phrase Topic Matching. We maintain a curated taxonomy of twelve social conflict topics (Table 2). Each topic is defined by a name, a one-sentence description, and a list of seed phrases. Matching uses whole-word regular expressions that tolerate common morphological variants (e.g., *redlin* matches *redlining*, *redlined*, *redlines*). Multiple topic assignments are permitted.

Topic	Example seed phrases
Policing & Public Safety	police brutality, use of force, qualified immunity
Voter Suppression	voter id, gerrymandering, polling place closures
Book Bans & Anti-DEI	book ban, critical race theory, dei ban
Housing & Displacement	eviction, gentrification, affordable housing
Maternal Health	maternal mortality, birth equity, obstetric racism
Redlining & Fair Housing	redlining, discriminatory lending, fair housing
Anti-Black Surveillance	facial recognition, predictive policing
Reparations	reparations, slavery redress, atonement fund
School Funding	school funding, education funding, title i
Criminal Justice Reform	mass incarceration, mandatory minimum, bail reform
Environmental Justice	environmental racism, cancer alley, sacrifice zone
Economic Equity & Wealth Gap	racial wealth gap, predatory lending

Table 2: Social conflict topic taxonomy (12 topics).

3.3 Stage 3: Report Generation

A Jinja2 HTML template is populated with the analysis outputs and rendered to PDF via WeasyPrint [11]. The report includes: (i) a per-source article count and data quality table, (ii) a topic frequency bar chart, (iii) a source-by-topic heatmap, (iv) per-topic TF-IDF keyword word clouds, and (v) an annotated article index with clickable URLs grouped by topic. Chart rendering uses matplotlib and seaborn.

3.4 Stage 4: Delivery & Automation

Output files (combined CSV, seven per-source CSVs, HTML and PDF reports) are uploaded to Google Drive via the Drive v3 REST API with OAuth 2.0 authentication. Existing files are updated in-place rather than duplicated. The pipeline is fully automated via GitHub Actions (ubuntu-22.04), running on the first of every month via a scheduled cron trigger. It also supports a `workflow_dispatch` event, which allows any collaborator with repository access to trigger a full pipeline run manually at any time—useful for reprocessing after a scraper fix, testing a new topic addition, or producing an ad-hoc report outside the regular monthly cadence. End-to-end wall-clock time is approximately 5–8 minutes.

4 DATASET CHARACTERISTICS

4.1 Coverage

In testing over the February–March 2026 window, the pipeline collected 1,218 articles across the seven sources (Table 3). Article counts per source vary significantly, reflecting both publication cadence and the diversity of scraping methods required.

Source	Articles	Date%	Snippet%	Cat.%
thegrio.com	389	100	97	98
theroot.com	312	100	95	95
newsone.com	278	100	93	91
ebony.com	84	100	88	72
essence.com	71	96	82	68
capitalbnews.org	51	100	91	85
blavity.com	33	88	79	61
Total	1,218	99	93	90

Table 3: Approximate monthly article yield per source.

4.2 Schema

Each article record contains eight fields: title, description, source, date_of_publication (YYYY-MM-DD), category, author, link, and topics (the last field appears only in `articles_tagged.csv`).

4.3 Topic Distribution

Across the test window, **Policing & Public Safety** (18%), **Criminal Justice Reform** (14%), and **Economic Equity & Wealth Gap** (12%) were the most frequently matched topics. **Reparations** (3%) and **Anti-Black Surveillance** (4%) had the lowest match rates, consistent with their more targeted seed-phrase sets. Approximately 31% of articles received no topic match, indicating non-conflict-focused content (culture, sports, entertainment) or articles whose match tokens appear only in the body text.

5 LIMITATIONS & FUTURE WORK

5.1 Scraping Reliability

JavaScript-rendered sites require brittle workarounds. Three outlets—Blavity, Essence, and Capital B News—do not expose clean server-rendered HTML or functional WordPress REST APIs. Blavity’s sitemap returns HTML not XML; publication dates are injected by client-side JavaScript and recovered only via application/ld+json blocks; internal links use root-relative paths requiring explicit base-URL prepension; category URLs follow /categories/X not /category/X. Each issue independently produced zero valid articles during development.

Anti-crawling measures suppress yield. Blavity’s CDN occasionally serves challenge pages to automated clients. Capital B News exposes over 1,000 candidate URLs; the scraper caps at 200 and limits pagination to 3 pages to keep runtime under 5 minutes.

Scraper fragility under site redesigns. HTML scrapers are coupled to DOM structure. WordPress REST API scrapers are comparatively robust. We plan to add per-source yield regression tests that alert when any source returns significantly fewer articles than its historical baseline.

5.2 Coverage Asymmetry

Article yield varies substantially: The Grio and The Root contribute ~300–400 articles per month each; Blavity ~30–35 and Capital B News ~50. Aggregate TF-IDF scores and topic frequencies are therefore dominated by higher-volume outlets. Researchers conducting cross-outlet analysis should weight or stratify by source before drawing community-level conclusions.

5.3 Topic Annotation Quality

Surface-text matching only. The matcher operates on title and meta-description fields only. ~31% of articles received no topic match, reflecting both off-topic content and cases where topic terms appear only in the article body. A planned extension will optionally store and match against extracted body text.

Label boundary overlap. A story about school segregation in a redlined neighbourhood may match both *School Funding* and *Redlining & Fair Housing*. Future work will evaluate zero-shot LLM classification using topic descriptions as natural-language hypotheses (e.g., via an NLI head), producing confidence-calibrated multi-label annotations.

Towards a fully agentic pipeline. Beyond improving individual annotation steps, we plan to develop an end-to-end agentic process in which LLMs drive both data collection and analysis. In this architecture, an LLM agent would (i) discover new publication sources relevant to a user-specified community or topic domain, (ii) generate and iteratively refine scraper code for those sources, (iii) synthesise topic ontologies grounded in the collected corpus rather than pre-specified seed phrases, and (iv) produce natural-language analytical summaries alongside the structured outputs. This would transform the pipeline from a fixed-topology ETL system into a self-configuring research assistant capable of targeting arbitrary underrepresented media ecosystems with minimal human intervention.

No sentiment or framing analysis. The pipeline identifies *what* topics are covered but not *how* they are framed. Extending the annotation layer with frame classifiers (e.g., using the Media Frames Corpus as a training signal) would substantially increase utility for framing studies.

5.4 Temporal and Structural Constraints

Rolling 30-day window. Each run retains only the preceding 30 days. Longitudinal analysis requires aggregating versioned outputs across runs—supported by the dated-subfolder structure but not automatically merged. A future stage will maintain a deduplicated cumulative corpus alongside per-run snapshots.

No article body storage. The schema stores only titles and meta-descriptions to minimise storage and avoid copyright concerns. We are exploring a lightweight opt-in body extraction mode gated behind explicit fair-use acknowledgement.

5.5 Extensibility to Other Ecosystems

The modular design—isolated scraper modules, shared schema normaliser, declarative publication registry—supports extension to Indigenous press, Spanish-language outlets, LGBTQ+ media, and diaspora publications. The topic taxonomy is fully configurable through `analysis/topics.py` with no other changes. However, onboarding a new outlet requires manual investigation of its technical stack—a process that took 30 minutes for WordPress outlets and several days for Blavity.

6 CONCLUSION

We have described an open-source pipeline that addresses a concrete gap in computational journalism and social-computing research: the absence of continuously updated, community-press datasets annotated against sociologically grounded topic taxonomies. The system delivers approximately 1,200 research-ready articles per monthly cycle from seven Black media outlets, annotated across twelve social conflict topics, with no recurring manual intervention. We believe the architecture—modular scrapers, declarative publication registry, extensible topic taxonomy, CI/CD scheduling—serves as a replicable template for researchers who need live, domain-specific datasets for communities whose media presence is currently invisible to mainstream corpora. Source code, dataset samples, and documentation are available at [repository URL redacted for review].

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